

Merrivoss Bioanalysis Inc

CVL oral discovery package: microsomes, hepatocytes and plasma binding

Final Report

Study MV-2016-053 (LIMS M16/BIO/01247)

Sponsor: Corvenlea Ltd

Study Director: Dr M. Larkin

Report date: May 9, 2016

1 Summary

Forty compounds from the Corvenlea aminopyrimidine series were profiled for Corvenlea Ltd in Microsomal CLint (human and rat liver microsomes), Hepatocyte depletion CLint and fu,p by equilibrium dialysis. Matrix acceptance criteria were met in every run. Rat microsomal clearance was higher than human for most of the set, so rodent exposure is likely to under-represent human exposure for this chemotype.

Table 1 Study summary

Metric	Value
Project	MV-2016-053
Client	Corvenlea Ltd
Compounds tested	40
Series / chemotype	Corvenlea aminopyrimidine series 01 (Aminopyrimidine)
Assays	Liver microsomal stability, Hepatocyte stability, Plasma protein binding (equilibrium dialysis)
Turnaround (calendar days)	25
Price	USD 23,500

2 Test articles and matrices

Test articles were received on April 14, 2016 as 10 mM DMSO solutions in one accession lot. Matrices: pooled human and Sprague-Dawley rat liver microsomes (0.5 mg/mL protein); cryopreserved human and rat hepatocytes (0.5×10⁶ viable cells/mL, viability ≥80% at thaw); pooled human plasma (K2EDTA).

Analytical limits: LC-MS/MS with compound-specific transitions; lower limit of quantitation 1 nM in incubation matrix. Depletion below 20% over the incubation is reported as a lower bound.

3 Methods

- Microsomal CLint: 1 µM, NADPH, 0-45 min, five time points; CLint from the first-order depletion slope

- Hepatocyte depletion CLint: 1 μ M, 0-120 min, six time points; scaled per 10^6 cells
- fu,p by equilibrium dialysis: 5 μ M, 4 h at 37 °C; mass balance recovery reported

Controls: verapamil and diltiazem (microsomes), 7-ethoxycoumarin (hepatocytes), warfarin (plasma binding). All within historical ranges.

4 Results

Microsomal results are given in Table 2. Hepatocyte clearance (Table 3) ranked compounds in broadly the same order as microsomes, with the most lipophilic analogues cleared fastest in both systems. Plasma binding was moderate to high across the set (Table 4).

Table 2 Microsomal intrinsic clearance and half-life

Compound	Human CLint (μ L/min/mg)	Human $t_{1/2}$ (min)	Rat CLint (μ L/min/mg)	Rat $t_{1/2}$ (min)
CVL-08800	25	56	33	41
CVL-08810	10	>60	15	>60
CVL-08812	14	>60	22	>60
CVL-08816	4.4	>60	11	>60
CVL-08822	19	>60	30	47
CVL-08828	10	>60	14	>60
CVL-08830	13	>60	16	>60
CVL-08834	11	>60	25	56
CVL-08836	36	38	55	25
CVL-08840	4.4	>60	6.4	>60
CVL-08842	11	>60	34	40
CVL-08848	42	33	89	16
CVL-08852	9.4	>60	21	>60
CVL-08854	3.2	>60	9.3	>60
CVL-08860	19	>60	16	>60
CVL-08864	11	>60	22	>60
CVL-08872	6.9	>60	14	>60
CVL-08884	11	>60	12	>60
CVL-08888	9.9	>60	22	>60
CVL-08912	29	49	53	26
CVL-08918	17	>60	45	31
CVL-08920	6.0	>60	9.2	>60
CVL-08924	25	56	35	40
CVL-08926	8.1	>60	14	>60
CVL-08930	12	>60	34	40
CVL-08932	18	>60	36	39
CVL-08936	7.1	>60	12	>60
CVL-08942	5.9	>60	12	>60
CVL-08948	47	29	61	23
CVL-08954	9.5	>60	11	>60
CVL-08956	22	>60	30	46
CVL-08962	10	>60	6.9	>60
CVL-08966	33	43	64	22
CVL-08968	65	21	110	13
CVL-08972	23	>60	92	15
CVL-08974	11	>60	17	>60
CVL-08978	18	>60	27	52
CVL-08980	7.5	>60	14	>60
CVL-08984	27	51	66	21
CVL-08990	22	>60	23	59

Controls: verapamil (high), diltiazem (medium) within historical range

Table 3 Hepatocyte intrinsic clearance

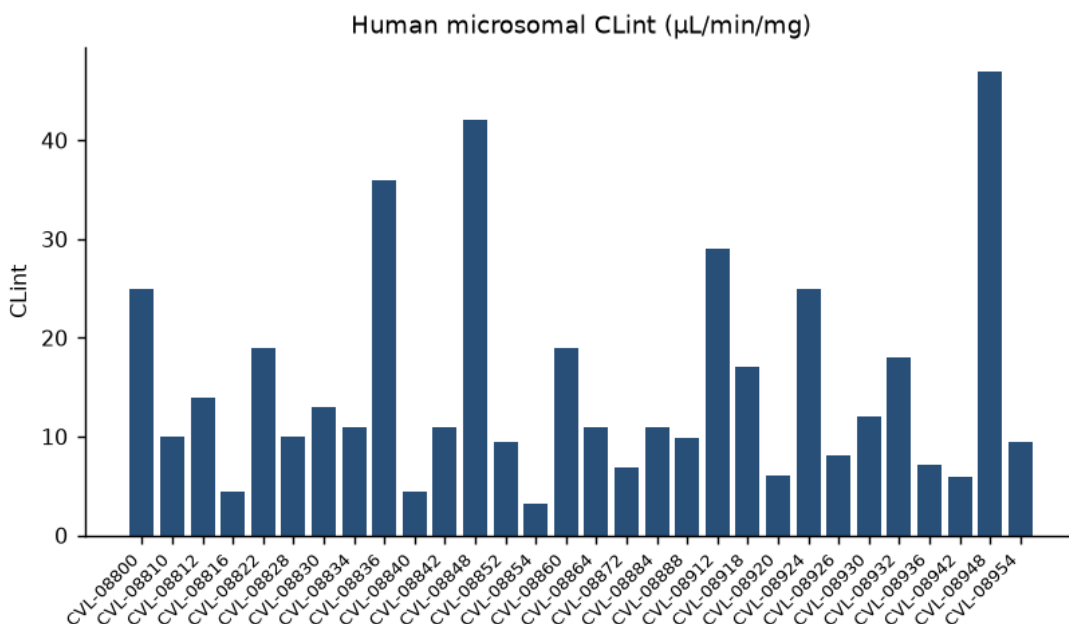
Compound	Human CL _{int} (µL/min/10 ⁶ cells)	Rat CL _{int} (µL/min/10 ⁶ cells)
CVL-08800	9.7	15
CVL-08810	10	13
CVL-08812	23	44
CVL-08816	36	67
CVL-08822	22	46
CVL-08828	15	26
CVL-08830	22	29
CVL-08834	12	44
CVL-08836	9.1	11
CVL-08840	16	24
CVL-08842	11	21
CVL-08848	38	60
CVL-08852	19	17
CVL-08854	11	17
CVL-08860	14	20
CVL-08864	28	67
CVL-08872	16	21
CVL-08884	7.5	13
CVL-08888	13	28
CVL-08912	10	17
CVL-08918	7.4	12
CVL-08920	13	33
CVL-08924	40	86
CVL-08926	21	30
CVL-08930	18	48
CVL-08932	15	40
CVL-08936	13	22
CVL-08942	11	17
CVL-08948	45	87
CVL-08954	30	40
CVL-08956	19	44
CVL-08962	22	29
CVL-08966	16	30
CVL-08968	17	29
CVL-08972	14	25
CVL-08974	7.0	14
CVL-08978	11	19
CVL-08980	31	89
CVL-08984	7.3	16
CVL-08990	8.4	24

Table 4 Human plasma protein binding

Compound	% bound	fu	Recovery (%)
CVL-08800	92.0	8.01	103
CVL-08810	91.8	8.18	92
CVL-08812	94.0	5.99	85
CVL-08816	79.5	20.50	97
CVL-08822	82.5	17.52	85
CVL-08828	88.9	11.06	83
CVL-08830	99.9	0.10	104

CVL-08834	97.4	2.57	100
CVL-08836	99.9	0.10	99
CVL-08840	80.8	19.17	103
CVL-08842	81.8	18.19	89
CVL-08848	89.7	10.32	103
CVL-08852	93.4	6.56	83
CVL-08854	79.7	20.31	92
CVL-08860	89.8	10.18	101
CVL-08864	98.0	1.98	85
CVL-08872	96.0	3.99	86
CVL-08884	88.3	11.68	105
CVL-08888	92.4	7.65	87
CVL-08912	91.3	8.68	99
CVL-08918	99.9	0.10	104
CVL-08920	89.6	10.37	99
CVL-08924	99.0	0.96	90
CVL-08926	99.9	0.10	94
CVL-08930	89.9	10.13	83
CVL-08932	89.1	10.86	81
CVL-08936	87.3	12.65	88
CVL-08942	92.1	7.93	85
CVL-08948	99.8	0.24	92
CVL-08954	87.1	12.86	102
CVL-08956	91.7	8.27	98
CVL-08962	99.9	0.10	86
CVL-08966	99.9	0.10	98
CVL-08968	99.9	0.10	104
CVL-08972	89.0	11.02	90
CVL-08974	89.6	10.42	86
CVL-08978	99.9	0.10	84
CVL-08980	81.0	18.98	103
CVL-08984	99.2	0.83	86
CVL-08990	91.2	8.83	82

Warfarin control 98-99% bound



5 Discussion

Exposure predictions should not be drawn from rat data alone for this series. Because rat clearance exceeds human for most compounds, a short rodent PK study on two or three representative analogues is recommended to calibrate in vitro-in vivo scaling before the next analogue set is chosen.

Free fraction data are reported for human plasma only; rat plasma binding was outside the agreed scope.

Name	Role	Signature	Date
Megan Larkin	DMPK Study Director	<i>Larkin</i>	May 9, 2016
Helen Marchwood	Bioanalytical Quality Manager	<i>Marchwood</i>	May 9, 2016